

B.Tech. Examination-2023-24
Department of Electronics and Communication Engineering
(Odd Semester Regular & Supplementary)
Course Title: Microprocessor and Its Application **Course Code: ECEUGOE04**

Full Marks: 80**Time: 3.00 Hrs**

- Answer all parts of a question in same place.
- Figures on the right hand side margin indicate full marks.
- Symbols have their usual meaning

Answer any Eight (8) Questions

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| 1. | (a) | Write down the differences between microprocessor and microcontroller. | 4 |
| | (b) | Why the lower byte addresses bus (A0-A7) and data bus (D0-D7) are multiplexed? What is the function of ALE pin of 8085 microprocessor and explain with the help of proper diagram. | 6 |
| 2. | (a) | Draw the timing diagram of MOV A, M (Opcode: 7E) instruction stored from memory location 8000H. | 5 |
| | (b) | Write a program to generate a 10msec time delay considering the operating frequency of 8085 is 4MHz. | 5 |
| 3. | (a) | What do you mean by maskable and non maskable interrupts of 8085? Write a program to enable all interrupt except RST 5.5 | 7 |
| | (b) | Write a program to exchange the contents of DE register pair with the HL register pair using PUSH and POP instruction. | 3 |
| 4. | (a) | Explain the flag structure of 8085 microprocessor with suitable example. | 6 |
| | (b) | Write down the function of the following instructions
a) ADI b) LHLD c) CMP d) XCHG | 4 |
| 5. | | Write an 8085 assembly language program to transfer the six 8-bit data is stored from memory location C070H to new memory locations starting from C050H. | 10 |
| 6. | (a) | Explain the function of HOLD, RESET OUT, READY signal and PC register of 8085 microprocessor. | 4 |
| | (b) | Draw and explain the Direct Memory Access (DMA) operation. | 6 |
| 7. | (a) | Interface the 8KB RAM, and 4KB EPROM memory devices in 8085 microprocessor in absolute decoding method. | 5 |
| | (b) | Design a seven-segment LED output port with the device address F5H, using a 3-to-8 decoder, a 4-input NAND gate, a NOR gate and a common anode seven segment LED. | 5 |
| 8. | (a) | Draw and explain the functional block diagram of 8255. | 5 |
| | (b) | Explain the BSR mode of 8255. Write a control word for PC7 set and PC5 reset. | 5 |
| 9. | (a) | Discuss the different bits of the control word of 5255. | 5 |
| | (b) | Write a control word for mode 0 to configure port A and port CU as output port and port B & port CL as input port. | 5 |

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| 10. | (a) | Draw and discuss the architecture of 8086 microprocessor. | 6 |
| | (b) | How do pipelining achieved in 8086 microprocessor. | 4 |
| 11. | (a) | Explain the process of segmentation of memory in 8086 microprocessor. What are its advantages? | 5 |
| | (b) | Draw and explain the flag register of 8086 microprocessor. | 5 |
| 12. | (a) | Write down the differences between 8086 microprocessor and 8088 microprocessor. | 4 |
| | (b) | Write an 8085 assembly language program to compare two strings. Assume that the first byte of both strings contain the number of bytes in that string. The starting address of the two strings is 9000H and 9100H respectively. If both strings are found equal place 11H in memory location 9500H else place 22H. | 6 |

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